

SQL CHATBOT-USING CONTEXT FREE GRAMMER

A Mini Project report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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(Affiliated to JNTU Hyderabad & Approved by AICTE New Delhi)

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March 2026

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CERTIFICATE

This is to certify that the project report entitled “**SQL CHATBOT-USING CONTEXT FREE GRAMMER**” submitted by the following students in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in CSE (Data Science) and is a Bonafide record of the work performed by.

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ACKNOWLEDGEMENT

We wish to pay our sincere thanks **Dr. CH. SRINIVAS**, Principal, Vaageswari College of Engineering, for providing all required facilities and his support during the mini project work.

We would like to thank **Dr. N. CHANDRAMOULI**, Professor and HOD of the Computer Science and Engineering Department for his valuable suggestions during the mini project work.

We sincerely extend our thanks to the Mini project guide **DR. MD. SIRAJUDDIN**, Associate Professor of Computer Science and Engineering Department, for sparing his valuable time in guiding the Mini project work and giving feedback with many useful suggestions during the major project work.

We are also conveying our heartfelt thanks to the Institute authority, Department, Library, and Laboratory staff of Vaageswari College of Engineering for their co-operation during our Mini project. We thank our beloved friends for their help and encouragement regarding the concepts and experimentation.

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ABSTRACT

In this project work, we derive the semantics of a given English query and convert it into its equivalent SQL query. Instead of using neural networks for semantic analysis of English queries, we opt for a Context Free Grammar approach. Most neural network-based systems can handle only one semantic at a time, whereas, because of the flexibility offered by our CFG approach, our system manages to handle simultaneous usage of conjunctive, disjunctive, and negative semantics. It also handles complex statements comprising of main as well as dependent clauses. In addition, the system also takes into account aggregate functions and constructs the required GROUP-BY and HAVING clauses. We describe how the system analyzes English queries by understanding the role that each part-of-speech has to play in constructing SQL queries. Numerous examples demonstrate the effectiveness of our approach where state-of-the-art techniques relying on deep learning algorithms fail to deliver.

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